Explaining a classifier's decisions

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Abstract

As complex classifiers become increasingly prevalent in sensitive domains, the ability to understand and trust the decisions of these black-box models is essential. This poster provides a comprehensive overview of current explainable AI (XAI) methods designed to interpret classifier outputs. First, logic-based methods attempt to formalize the classifier's reasoning to offer formal guarantees on the quality and rigor of the produced explanations. Second, Shapley value methods evaluate feature importance by assigning a contribution score to each input based on game theory principles. Finally, perturbation methods observe model behavior by generating perturbed input samples. These methods provide local explanations either through linear approximations or user-friendly IF-THEN rules.

1 Logic-based Abductive Explanations

Logic-based XAI use formal mathematics to prove the "why" behind each decision. In this part we will introduce the main branch of logic-based XAI that use AXp.

A **weak Abductive Explanation** (wAXp) corresponds to a set of feature values that guarantee the prediction, regardless of the values of the remaining features. It is formalized as a partial assignment E of an instance $x \in F$ such that : $\forall y \in F,\ E(y) \Rightarrow f(y) = f(x)$. Intuitively, an **Abductive Explanation** (AXp) is a subset minimal wAXp.

The following example shows a very simple classifier for an input x with two binary features $v1$ and $v2$: $f(x) = v1 \wedge v2$.

	v_1	v_2	$f(x_i)$
x_1	0	0	0
x_2	0	1	1
x_3	1	0	1
x_4	1	1	1

The decision $f(x_4)$ has three wAXp: $E_1 = \{(v_1, 1)\}$, $E_2 = \{(v_2, 1)\}$ and $E_3 = \{(v_1, 1), (v_2, 1)\}$ but only E_3 is a AXp. The decision $f(x_1)$ has one wAXp/AXP: $E_4 = \{(v_1, 0), (v_2, 0)\}$.

In this section, we present three different methods to find AXps.

Coverage-based Explanations

Standard AXp's assume **feature independence**. When dependencies exist between features (e.g. a pregnant person is necessarily a woman), AXp's produce superfluous or redundant explanations [1]. CPI-Xp solves this by incorporating **feature constraints** and selecting explanations with **maximal coverage**.

Let $F[\mathcal{C}]$ be the set of feasible instances under constraints $\mathcal{C}$, and $cov(E) = \{x \in F[\mathcal{C}] \mid E(x)\}$ the coverage of a partial assignment E. A **CPI-Xp** of $f(x)$ is a partial assignment E such that:

1. $E(x)$
2. $\forall y \in F[\mathcal{C}],\ E(y) \Rightarrow f(y) = f(x)$
3. No valid explanation E' strictly subsumes E (i.e. $cov(E) \subsetneq cov(E')$)

The preferred variant **pCPI-Xp** keeps one representative per coverage class and is the only notion satisfying all seven formal properties (success, non-triviality, irreducibility, coherence, consistency, independence, non-equivalence).

Complexity: Π_2^P-complete (exact), but a **polynomial approximation** exists via dataset-based versions (d-CPI-Xp) in $O(m^2 n^2)$.

Efficient and Rigorous Model-Agnostic Explanations

Logic-based XAI addresses the lack of rigor of sub-symbolic methods but has two major drawbacks, the need to access the structure of the model and the scalability problems. With efficient and rigorous model-agnostic explanations both problems can be solved. To do that, this model will search the **sample based abductive explanation** (sbAXp). To find the sbAXp, this model uses **weak contrastive explanation** (sbWCXp), which is a subset of features that suffice to change the prediction if their value is changed.

A Contrastive Explanation (CXp) is a subset-minimal WCXp,whereas an AXp answers a "Why?" question, a contrastive explanation answers a "Why not?" question. This XAI uses the duality of AXp and CXp explanations, meaning, from sbCXp we can get the sbAXp.

The algorithm to go from the subset of all sbWCXp to one sbAXp is quite simple: compare each sbWCXp against all others and keep the ones with no subsets to get all sbCXps. Finding one sbAXp can be done in polynomial time but finding one smallest sbAXp is much harder and unlikely to be solved in polynomial time, but can be done with a MaxSAT solver.

Leveraging Argumentation for Generating Robust Sample-based Explanations

This sample-based XAI uses **arguments**, which is a plausible sbAXp. To get the final sbAXps, this XAI uses attacks : if two arguments contradict each other, then both will be compared and only one (if any) will be kept, this is repeated until there is no contradiction.

Each argument is composed of two parts, first **the support**, which the condition of the argument, when does it apply, second is **the conclusion**, what class does the argument predict. A contradiction is when the union of each support is not null and the conclusion different. All attacks are treated at the same time and depending on the selection function will give a set of stable extensions. A stable extension is a set of arguments without contradiction. The choosing of one of the stable extensions is done with an inference rule.

With this XAI both success and coherence is not guaranteed. Coherence is only guaranteed with the universal inference rule, thus prioritize universal inference, keeping arguments that are in all selected stable extensions

2 Shapley values based XAI

This family of XAI methods are based on the Shapley values concept from game theory that give an **importance score** to each feature in the prediction of a single instance. Shapley values provide a mathematically fair and unique method to attribute the payoff of a cooperative game to the players.

Let $\mathcal{F} = \{1, \ldots, n\}$ be the set of the numbered features (the players). The Shapley value of a feature i is computed by going through all the coalitions not containing i and looking at the **payoff difference** with and without the feature i involved [2] :

$$\phi_i = \frac{1}{n} \sum_{\mathcal{S} \in \mathcal{F} \setminus \{i\}} \binom{n-1}{|\mathcal{S}|}^{-1} \left(\mathbb{E}[f(X) | X_{\mathcal{S} \cup \{i\}} = x_{\mathcal{S} \cup \{i\}}] - \mathbb{E}[f(X) | X_{\mathcal{S}} = x_{\mathcal{S}}] \right)$$

From the efficiency axiom of Shapley values [3], we get the following property that states that the sum of the importance is equal to the actual predicted value minus a reference contribution that is the mean of all the predictions for the distribution on the input X

$$\sum_{i \in \mathcal{F}} \phi_i(v) = \mathbb{E}[f(X) | X_{\mathcal{F}} = x_{\mathcal{F}}] - \mathbb{E}[f(X) | X_{\emptyset} = x_{\emptyset}] = f(x) - \mathbb{E}[f(X)]$$

Figure 1 vizualises computed importance scores for the prediction of a naive Bayes classifier on an instance of the Titanic dataset.

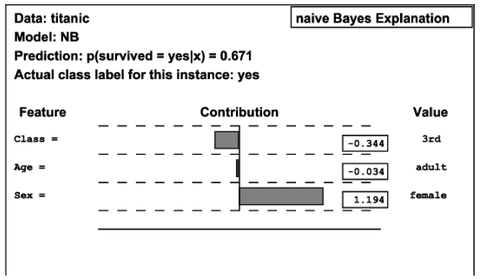

Figure 1: Feature importance attributions computed with Shapley values for the decision of a naive Bayes model on an instance of the Titanic dataset. From [4]

Approximation

Strumbelj and Kononenko [4] have shown that computing exact Shapley values become **intractable** when the number of features is large ($n \geq 100$). Merrick and Taly [2] have come up with sampling methods to get an approximation in polynomial time:

- sub-coalitions are sampled by taking all the features preceiding i in random orderings O_j of $\mathcal{F}$: $\mathcal{S}_j = Pre_i(O_j)$;
- the reference instances X are sampled following the distribution of input instances.

3 Perturbation model-agnostic method

Perturbation methods identify which input features most significantly influence a black-box model's decision process. These model-agnostic techniques compare the model's output for an original instance against its predictions for perturbed samples. By observing how these modifications affect the classification, we can quantify the importance of each feature without accessing the model's internal architecture.

LIME

LIME (Local Interpretable Model-Agnostic Explainer) generates local surrogates to approximate a black-box model f [5].

Key Principle: It minimizes a loss function $\mathcal{L}(f, g, \pi_x)$ to ensure local fidelity while keeping the explanation model g simple enough ($\Omega(g)$) for humans. A most global explanation of the black-box model can be provided by SP-LIME which select a set of local and non-redundant LIME explanations.

Limitation: The explanation depends heavily on the kernel width π_x and sampling, leading to potential instability.

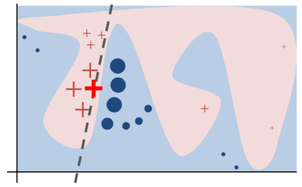

Figure 2: LIME linear local approximation. blue circles: f classes. red cross: explaned instance. brown crosses: perturbation sample. dotted line: local approximation of g.

Anchors

Anchors were developed to solve the lack of clear boundaries provided by LIME. To perform that, Anchors generates IF-THEN rules which 'anchor' a prediction. This set of short and disjoint rules are more comprehensible and then preferred by the users [6].

Formal Definition: A rule A is an Anchor for x if $A(x) = 1$ and it maintains high precision τ among its neighbors $\mathbb{E}_{\mathcal{D}(z|A)}[1_{f(x)=f(z)}] \geqslant \tau$, with $\mathcal{D}(x \mid A)$ the distribution of neighbors of x which satisfy the Anchor A.

Computing Anchors is a combinatorial optimization problem. To handle the exponential search space, a **Bottom-Up approach** is used:

- **Candidate Generation**: Rules are extended iteratively with additional predicates.
- **Selection via MAB**: A Multi-Armed Bandit (*KL-LUCB*) algorithm identifies the most precise candidates using minimal model calls [6].
- **Beam Search**: To avoid purely greedy results, the algorithm maintains B candidates, finally outputting the one with the **highest coverage**: $cov(A) = \mathbb{E}_{\mathcal{D}(z)}[A(z)]$.

References

[1] Martin Cooper et al. Sept. 2024. **[2]** Luke Merrick et al. en. In: ed. by Andreas Holzinger et al. Cham: Springer International Publishing, 2020. **[3]** Daniel Fryer et al. In: *IEEE Access* 9 (2021). **[4]** Erik Strumbelj et al. In: *J. Mach. Learn. Res.* 11 (Mar. 2010). **[5]** Marco Tulio Ribeiro et al. en. In: San Francisco California USA: ACM, Aug. 2016. **[6]** Marco Tulio Ribeiro et al. In: vol. 32. 1. 2018.